

Points to Consider for Design and Control of Continuous Bioprocessing

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36.1 INTRODUCTION

Continuous and semicontinuous processes have been practiced in many industries for decades. Prime examples are steel, paper, and automobile production. These industries are all very capital intensive and thus, each has switched to continuous processing as a primary mode of production, as it is considered the most cost efficient engineering implementation for manufacturing. Continuous operation can help in maximizing equipment utilization. Consequently, for equal production capacity, flow rates of process streams are smaller, equipment size is reduced, and facilities are smaller.

In small molecule enantiomer production, continuous separations have been performed in the form of simulated moving bed chromatography [1–3]. In the last decade, a renewed interest has emerged in the development and implementation of continuous processes for chemical active pharmaceutical ingredients. Several companies have made significant investments in continuous processing lines and established academic collaborations to advance enabling technologies [4,5].

Alternatively, in the production of therapeutic biological products, companies have been extremely cautious towards adoption of continuous processing. Until now, implementation has been limited to continuous perfusion cell-culture processes, mainly for products that are unstable in the bioreactor. This hesitancy is caused in large part by unwillingness to confront regulatory uncertainties and the necessary changes to established quality systems due to the large installed batch production capacity. Recently though, interest in continuous manufacturing of therapeutic biological products is increasing [6–10]. Continuous processing is a promising answer to pressures on the industry of how to increase productivity, and to reduce costs. Several potential advantages have been evaluated for their realization potential [11,12].

Developing a fully integrated continuous process for a biological product first requires the conversion of individual unit operations to continuous operation, and second, the combination of those unit operations into one integrated process [13]. The highly regulated nature of the production of biological therapeutics creates unique points to consider. Continuous processes require a high level of process understanding and control. For that reason, highly robust and reliable methods for at-line process analytics and real time monitoring must be in place. Unlike quality control in many other industries, for biologic therapeutics there is a regulatory requirement that quality be built into the process rather than tested into the final product [14], as well as demonstration of consistent and controlled execution of the manufacturing process [15]. This elevates the importance of careful and controlled implementation of continuous processing and the importance of developing robust and reliable process analytical technologies (PAT) [16,17]. Fortunately, regulatory agencies have expressed their support for continuous processing due to its promise to lower drug costs, and are expected to be open-minded when reviewing regulatory submissions describing new continuous technologies [18]. Nevertheless, robust and reliable control of the product's safety and efficacy will always be the primary focus of regulators [19]. Numerous technological and operational challenges must be overcome to reach the level of understanding and operational robustness required for commercial implementation. In the following sections, some of these challenges will be discussed based around the example of the production of a monoclonal antibody. Monoclonal antibodies are products with large mass demands for the market, therefore considering production costs is required to produce cost effective therapies for patients. Monoclonal antibodies are typically very stable products and concerns about product residence time and stability are of a lesser concern. For products where product stability at certain process conditions is a concern, continuous processing can offer additional benefits that are not discussed here.

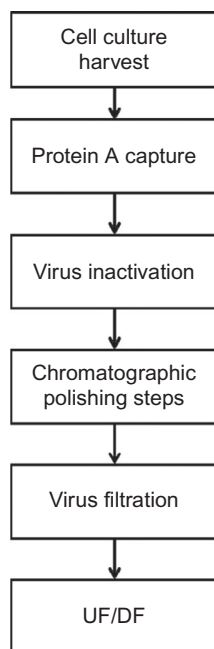


FIG. 36.1 Typical platform process for monoclonal antibodies.

A typical traditional production process [20] for a monoclonal antibody is outlined in Fig. 36.1 and starts with multiple cell culture steps designed to expand the cell mass of the culture from the small amount in a vial of a cell bank to a sufficient amount to seed a production reactor. In the production reactor, the product is formed under controlled conditions to allow for high productivity, robustness, and reproducibility. At the end of the production step the cells must be removed from the product; a step typically performed by centrifugation or membrane separation. The cell free fluid is loaded onto and eluted from a Protein A chromatography column that removes most impurities and concentrates the product. This capture step is followed by a low pH virus inactivation step to ensure virus safety of the product. The product is then usually further purified by one or more additional chromatography steps to ensure robust control of contaminants like dimerized or aggregated product, host cell protein contaminants, and DNA. To ensure additional virus safety of the product, these polishing steps are typically followed by a filtration step with a virus retaining filter, followed by a formulation step that ensures the final drug substance is at its target product concentration and contains the correct excipients and buffer components. This formulation step is typically performed by tangential flow filtration (TFF).

All steps of the traditional process are fundamentally batch operations and must be redesigned in order to be performed in a continuous or pseudo-continuous mode.

In the following discourse, individual unit operations and their conversion to continuous operation will be discussed, followed by consideration of their combination into a fully integrated continuous process. Finally, overarching aspects of bioprocessing like fault recovery, development timelines, and similar considerations will be examined.

36.2 DEVELOPMENT AND IMPLEMENTATION

There is limited consensus about the impacts on process development timelines and resource requirements for continuous processes. Evaluation outcomes vary from expectations of shorter development timelines due to more flexibility to longer development timelines due to the longer duration of individual experiments.

To understand impact on process development, capabilities of the particular organization must be evaluated carefully to develop specific projections based on reviews of a number of competencies. Considerations will likely include

- Experience with cell line stability over the extended cell culture durations.
- Status of automation for small-scale execution and sampling of continuous experiments.
- Availability of analytical support for real time experiment monitoring and PAT method development.
- Potential for real time data analysis of several parallel experiments.
- Effect of number and duration of individual experiments on development duration.
- Achievable efficiencies from the uninterrupted availability of material for testing and development.

At a minimum, process characterization and knowledge on flow velocity, load rate, and stream concentrations needs to be greater than what would typically be needed for a batch process. This is because the continuous process will be performed with varying flow rates to maintain a continuous feed while accommodating variability in product mass and stream volume from the preceding operation in the process stream. Unit operation loading and pooling will need to accommodate varying mass and purity of process streams as continuous flow at varying flow rate replaces intermittent flow of predefined flowrates. This will require extra resources and potentially time to create expanded process knowledge. Virus clearance validation approaches may have to be developed to support the continuous operations. A review of the necessary adaptations of virus clearance validation strategies to support continuous processing [21] outlines a path forward and where innovation is needed.

Process development timelines and resource requirements are only the first important aspect of transitioning to continuous processing.

Transferring and executing the process in a GMP manufacturing environment will also require significant short term resources to develop the necessary capabilities. These initial resources will be substantially eclipsed by the significant long term benefits. One persistent implementation cost that can be higher when operating in continuous mode compared to batch is the extended time of facility occupation for the execution of process performance qualification (PPQ) and validation. As a typical continuous perfusion cell culture process is performed for a significantly longer duration than a batch or fed-batch cell culture process, and facility occupation cost is typically a dominant factor of the overall production cost, the associated extended facility occupation must be considered. The associated lower capital outlay of smaller facilities and equipment can possibly offset that disadvantage. The amount of product generated in multiple PPQ batches can exceed the initial commercial product demand and result in product loss to scrap or expiry. This makes the choice of production scale and batch duration very important and favors the implementation of a continuous process for life-cycle management of an already marketed product with known product demand.

36.3 DESIGN OF UNIT OPERATIONS

36.3.1 Production Cell Culture

Continuous perfusion cell culture has been well established for many years in the production of biologics. In the past, this production mode was most desirable for labile products that benefit from immediate removal from the cell culture fluid. Recent advancements have made this production mode more attractive for high productivity applications [22]. The discussion will be limited to aspects of cell culture that affect subsequent downstream processing and process design. A critical factor for overall process efficiency is the consistency of product output both with respect to product mass and stream volume per time. Consistent product quality over the duration of continuous production is paramount, as only then can a continuous process integrate both upstream and downstream. If there is meaningful variability in product quality over the production duration, the cell culture product can be collected into a homogeneous pool before further processing, negating most of the benefit of a fully integrated continuous process, or continuous monitoring and control of in-process quality attributes must be established. However, there may still be some benefits in having a combined pool further downstream from the initial harvest stream, when the product is in a more concentrated and stable solution condition.

With consistent product quality, the focus can shift to the operational variabilities of the cell culture process. The main factors affecting consistency of mass and volume of the product stream from the reactor are cell retention and cell density control. Fig. 36.2 depicts the typical process stream for a continuous bioreactor. A primary objective of the continuous culture production is to retain all viable cells in the reactor for continued growth while simultaneously producing a clarified product stream that can be loaded directly on a capture operation, which for monoclonal antibodies is typically a Protein A capture chromatography step. The choice of cell retention and cell bleed technology has a significant effect on the product stream.

Currently, microfiltration is a very popular choice for cell retention. With this technology, usually substantial membrane fouling is observed such that one membrane cannot support the entire continuous process duration without declines in product sieving. Over time this leads to decreasing product concentration in the membrane permeate and product accumulation in the reactor [23]. If membranes are replaced over the course of the production duration, large swings in product concentration in the permeate stream must be expected; peaks when accumulated product is released through a new membrane and valleys towards the end of use of a fouled membrane (Fig. 36.3). The capture process design needs to decide if the column will be sized for the average cell culture output or if the column has to be able to capture all product, even during short term peak release of accumulated mass.

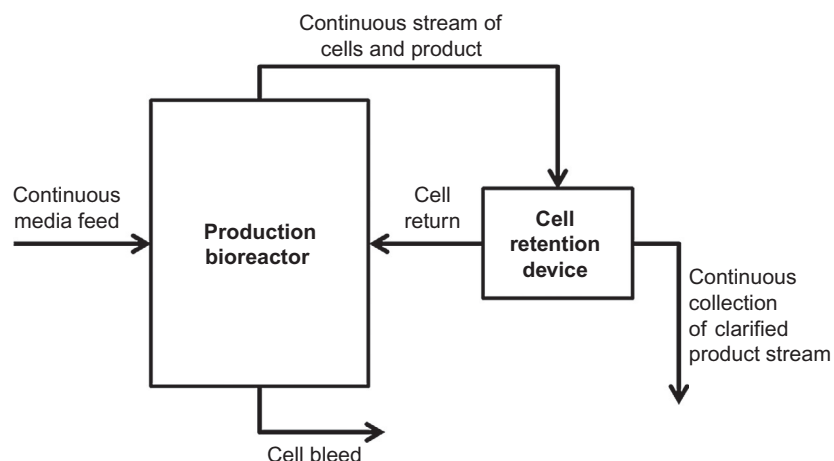


FIG. 36.2 Summary of process streams of a continuous bioreactor operation.

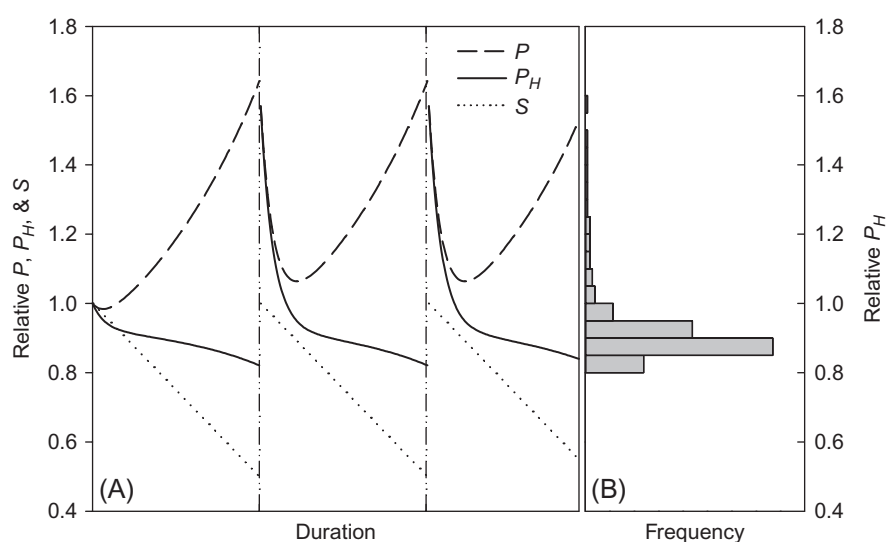


FIG. 36.3 Effect of product sieving (S) in a cell retention device with fouling. (A) Effect on product concentration in the reactor (P) and in the harvest stream (P_H). Solid line: P_H ; dashed line: P ; dotted line: S ; dash-dotted line: change-out of cell retention device. (B) Distribution of P_H .

Other cell retention technologies like inclined settlers [24], acoustic wave separators, or continuous centrifuges may avoid this particular challenge, but all these technologies are similarly challenged by the high cell densities that are targeted to achieve the high productivities desired.

Another key objective in continuous cell culture is maintaining consistent viable cell density in the production reactor. While the cell culture is targeted to minimize further cell growth and maximize product formation, there is always ongoing cell growth that needs to be counteracted with targeted removal of cells and cell debris. This cell removal is frequently accomplished with a small bleed of culture fluid containing both cells and product. Typically, product in the cell bleed stream is discarded. Even with perfectly consistent cell growth rates there must be an expectation of varying product loss in the bleed stream and consequently added variability in the amount of product harvested for further processing. Additionally, any variability in cell growth can also affect the volumetric output of the reactor unless the perfusion control strategy is designed to compensate for this variability. Usually though the reactor control strategy is focused on maximizing productivity and minimizing media requirements while also maintaining a stable cell culture and product quality.

36.3.2 Continuous Cycling Capture Chromatography

In monoclonal antibody production, the capture step is a very capital intensive operation due to high cost of Protein A resin. Alternative nonaffinity separation technologies like aqueous two-phase extraction [25] or traditional ion-exchange

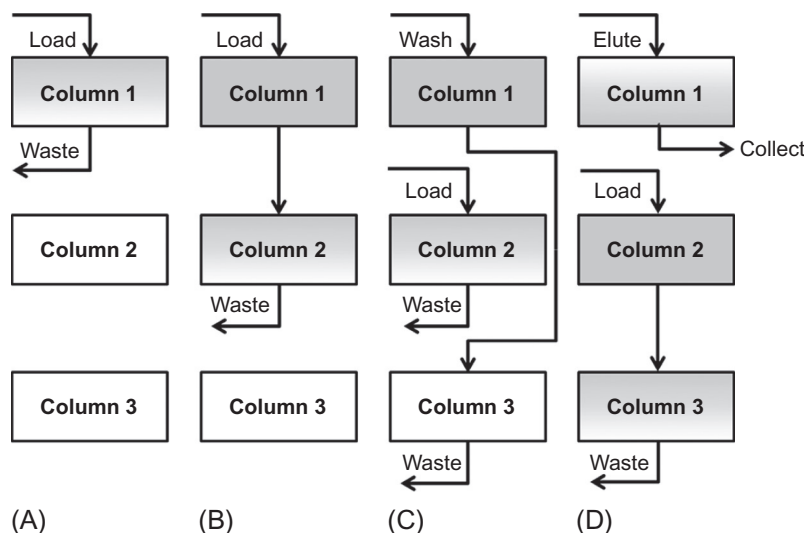


FIG. 36.4 The phases of a periodic column cycling operation. (A) Column 1 is loaded; (B) column 1 is overloaded and product breaking through is captured on column 2; (C) column 2 is loaded and column 1 is washed, unbound product is captured on column 3; and (D) column 1 is eluted and turned around for the next use, column 2 is overloaded and product breaking through is captured on column 3.

chromatography can be used [26] but have largely not been able to outperform the extremely high selectivity of a Protein A separation and the convenience and efficiency of being able to utilize one unit operation with only minor adjustments for multiple products. For that reason, recent focus has been on maximizing the productivity of the Protein A operation instead of replacing it outright [27]. There have been numerous studies and reports on the use of continuous cycling operations for a more efficient Protein A chromatography capture step [8,28–30]. The phases of a periodic column cycling operation are outlined in Fig. 36.4. Columns can be loaded beyond their capacity and any product breakthrough will be captured in a subsequent Protein A column while, simultaneously, a fully loaded column from the previous cycle is washed, eluted, and regenerated for its next cycle. The focus of this discussion is on process design and control aspects of these periodic cycling approaches. Alternative operating modes to packed chromatographic columns based on TFF systems have been proposed [31] to integrate harvest and capture processes into a periodic cycling operation. The general points to consider relating to operational robustness, design for feed variability, and resin and buffer savings are unchanged. While the term “column” is used in the following discussion, general principles will apply to periodic cycling approaches independent of format.

As highlighted in the previous section, there is an expectation of variability in volume and product concentration of the incoming load stream. However, there is also significant variability in the Protein A operation itself that needs to be accounted for. There should be an understanding that resin lot-to-lot variability and variability between column packs could affect the operation of multiple columns operating in concert in a periodic cycling operation. Yet the largest known variability that must be addressed in the design of a continuous chromatography process is the decline in resin capacity and process performance with extended lifetime use. Since increasing the number of resin reuses favorably counterbalances the high costs of Protein A resin, there must be robust controls and process understanding of process performance throughout the resin's lifetime. Similar to microfiltration as discussed in the previous section, the peaks and valleys over the lifecycle must be comprehensively understood and accounted for in the process design.

In traditional batch operation, the resin is loaded to a target level that has a safety factor from the real known binding capacity. This safety factor is chosen in combination with the maximum allowable resin use such that the performance decline over extended resin use is acceptable [32]. With periodic column cycling operations, the objective is to fully load one column before targeting the load stream to the next column in the sequence. This has multiple implications on process design and control. It has been shown that binding capacity is not necessarily a good measure of resin performance. Recoverable product can decline faster with extended use than a decrease in binding capacity due to increasing product loss during postload washes [33]. Increasing wash losses would be detrimental to the productivity of the periodic column cycling process design. In most cases, lower process yield will outweigh any benefits of resin cost savings. This problem can be overcome by collecting the product containing wash stream on another column in the cycling design, but it will negatively affect process efficiency by extending the nonloading duration for each column. There are potential implications to the removal of viruses and other impurities that must be considered in cases where the wash fraction is loaded onto a protein A column.

Another aspect of declining resin performance is that the system must be designed for resin performance at the last load cycle. The implications are manifold: If the system is designed to accommodate resin performance at the last use cycle and variabilities expected from the upstream process, the system must be significantly oversized for early resin use cycles and will only rarely operate at its capacity. During early resin use cycles the load duration can be significantly longer than required for parallel column turnaround of a nonloading column leading to columns waiting idle. Therefore, a design choice must be made between two fundamental objectives: robust simplicity or minimization of buffer use.

The first process design objective is having the simplest robust operation which always targets a nominal load volume. This allows for the simplest process control with predefined and never changing process operation. The consequence of never actually overloading columns is that the full benefits of resin and buffer saving cannot be achieved and the only benefit is in the continuous load stream integrated with the cell culture process.

The second process design objective is to always maximize column loading and in turn maximize resin and buffer savings. This requires some form of product detection in the effluent of the first column in the periodic cycling design. A solution utilizing multiple UV detectors with fully synchronized calibration has been proposed [13]. This approach requires more sophisticated operations planning and scheduling as it leads to varying load durations, buffer demand, cycle numbers and elution pools per day. When designing the subsequent process operations, variable product concentration over the reuse lifetime of the Protein A resin must be accounted for. Due to changes in resin capacity, the pool concentration from a column's first cycle will likely be higher than the pool concentration from a column's last cycle. Additionally, the aforementioned faster decline in recoverable capacity compared to binding capacity over the lifetime of the resin highlights the need for careful evaluation of step yields when using this approach to load to the maximum binding capacity.

For both approaches, there must be awareness that levels of host cell protein impurities have the potential to be negatively affected by overloading Protein A capture columns. In the first case, while maintaining the same load effectively the column only gets overloaded late in the resin use cycles. In the second case, the resin is loaded to its capacity in each cycle and there is little available information on how this operating mode affects product purity. In both cases this has to be overcome by comprehensive characterization work and by designing sufficient redundancy and robustness in subsequent unit operations to ensure overall process robustness and reliability.

Alternative operations have proposed the use of Protein A membrane adsorbers or monoliths for their better flow properties. This would give an advantage in the continuous cycling operation where low pressure drop and high flow rates are achievable and with favorable product yield. However, in most cases either the binding capacities or the wash out characteristics are insufficient. Poor wash-out characteristics when changing from one buffer to the next leads to increases in buffer volume requirements that are detrimental to large scale production [34]. Similar considerations apply to proposed operations of suspended affinity resin in tangential flow operations. Also, there, the buffer volumes required only allow for small scale operations.

36.3.3 Virus Inactivation

In most current biological therapeutic protein purification processes, there is a batch virus inactivation step. These batch inactivation steps are comprised of a low pH hold or a solvent/detergent treatment of the process stream for a defined incubation duration and temperature. Virus inactivation by low pH has proven to be an extremely robust virus clearance strategy. The low pH treatment is so robust and reliable that an ASTM standard has been developed that ensures at least 5 logs of xMuLV clearance [35]. To preserve the history and experience with this step, the simplest implementation for continuous manufacturing is to duplicate the inactivation equipment for two parallel operations. The tanks should be sized sufficiently to allow collection of an incoming process stream in one tank while the parallel tank performs all pH adjustment steps, including the hold duration, the feeding of the subsequent operation, and tank turnaround. This approach would not raise any regulatory concerns as it is robust and well defined. The duplicate tanks would each be significantly smaller than a tank utilized in typical batch operation. This semicontinuous approach is merely a duplication of traditional batch operations [36].

Several alternative approaches to continuous virus inactivation have been proposed that have not yet been proven at manufacturing scale or tested with regulatory agencies. These approaches will be briefly discussed.

One general approach is centered on combining the inactivation treatment with the capture column operation. Exposing the product to a low pH/high salt condition for a similar duration to traditional batch low pH inactivation while bound to the protein A resin provides clearance results comparable to the traditional approach [37]. This coupled operation can be implemented into the periodic column cycling capture operation. The negative of including virus inactivation into the column cycling operation is the additional 1–2 h of column hold duration required. Longer hold times during turnaround can significantly lower the productivity of the Protein A operation, unless the load concentration of the capture step is sufficiently low, making the load duration suitably long for coupling viral inactivation with the periodic capture operation.

Therefore, the objective for this approach must focus on minimizing the required hold duration for virus inactivation to maintain high productivity of the combined operations.

An alternative approach is to develop a true flow through reactor with control of a minimum residence time (MRT) at the target low pH or solvent detergent exposure condition. Coiled flow inverter designs have been shown by modeling as near plug-flow reactors, ensuring well defined MRT [38]. The theoretical understanding of this approach still needs to be validated through virus clearance studies. Ensuring sufficiently long exposure duration to compensate for uncertainties about MRT will define the overall required piping length of the reactor.

A methodology investigated for many years for its virus inactivation performance is ultraviolet-C (UV-C) irradiation [39]. This method is used in continuous mode in other applications but has not been widely used in biologic manufacturing due to the risk of product damage. Limitations of penetration depth have largely been overcome using spiral tube continuous reactors. Product specific studies will be required to understand and define the target UV-C exposure to ensure adequate virus inactivation while minimizing negative effects on product quality.

36.3.4 Continuous Polishing Chromatography

One or two polishing steps are typically performed to ensure robustness and control of removal of product and host cell related impurities. The traditional polishing steps are mostly ion-exchange or hydrophobic interaction chromatography. More recently, mixed mode steps combining both ion-exchange and hydrophobic interactions in single unit operations have become increasingly widespread. For the development of a continuous polishing operation the most important aspect is not necessarily the mode of chromatography, but rather the mode of operation: bind-and-elute (B&E) or flow-through (FT).

It is possible for B&E operations to be implemented in continuous mode, but they are much more difficult to implement than FT operations. B&E operations with step elution of the product can be implemented analogous to the Protein A operations discussed above, but the expectation of a significantly higher load concentration at this point of the process, due to the large concentration effect over the Protein A capture operation, makes the design much less productive. Load durations are likely very short and do not provide sufficient column turnaround time for periodic cycling. There are two approaches which could enable continuous operation: (1) an additional column in the turnaround phase to allow for faster readiness of the next load column, or (2) increasing the column diameter and slowing the load flow rate to provide sufficient turnaround time. Both approaches affect productivity of the operation negatively.

For B&E operations that require a linear gradient elution, a sophisticated multicolumn, countercurrent solvent gradient purification (MCSGP) approach has been developed [40,41]. Unfortunately, the aforementioned limitations of turnaround time are further amplified for this design. The linear gradient portion of the MCSGP process cycle adds even more turnaround time. However, the same two operational mitigations mentioned above can be applied. One attractive aspect of the MCSGP approach is the ability to reprocess product containing side fractions that did not satisfy product purity requirements. With that, the MCSGP method can improve the yield versus purity relationship, i.e., enable higher yield for a given target purity than what is possible in batch operation. On the other hand, applying this aspect of the technology will make the operation almost certainly not available for the overall virus clearance claim of the process. It will be practically impossible to perform an analogous continuous small-scale operation required for virus clearance validation.

FT operations are strongly preferred for implementation as a polishing step in continuous operation. The comparatively long load times and short turnaround times allow much simpler designs with two alternating columns. One column will be loaded to its normal load target while the second column is turned around for the next use. The FT operation also creates not just a continuous load flow but, with only minor interruptions, a continuous product output stream with constant concentration. Since the development of a continuous FT operation is identical to batch operation, FT process characterization and viral clearance validation can be performed with a small-scale batch study, utilizing a single column.

For continuous FT operations, alternative adsorber formats like membrane devices or monoliths can be attractive as they allow for faster flow rate. High binding capacity is usually not required in these cases as the membrane adsorbers bind low concentration impurities.

36.3.5 Virus Filtration

Filtration based virus clearance is a critical step for any biological therapeutic manufacturing process. In principle, they must be implemented very similarly to FT chromatography designs. These filtration steps are limited by fouling and load volumes that are determined in the small-scale validation studies. The difference versus the FT chromatography designs is that the filters are not regenerated and recycled, but swapped out for new units after each use. This causes the operation to be inherently less automated, although having oversized filters lessens the need for manual interaction. For a continuous

operation, typically the flow through of the filter will be maintained at a constant rate until a certain pressure threshold is reached. At this point, the flow rate must be lowered to maintain the pressure below the operating limit or flow has to be diverted to a parallel filter while the spent filter has to be changed. The frequency of change out could be minimized by using additional filtration area. The main equipment design challenge is the preservation of a closed system [21]. Environmental controls post virus filtration are typically very stringent and the repeated connection and disconnection of new filter units poses a significant risk to process integrity. Operational design must be focused on these aspects of process integrity, as well as operational and procedural controls.

36.3.6 Continuous Concentration and Formulation

In most biologics manufacturing processes, the purified product is formulated to its final concentration and into its final excipients by a TFF operation called ultrafiltration and diafiltration (UF/DF). In batch operation, the TFF process typically starts with a concentration phase (UF), where the product is retained and buffer passes the membrane as permeate. This is followed by a buffer exchange where the product concentration is held constant and excipient concentrations are defined in a DF step. This is finally followed by an UF overconcentration operation to allow product recovery from the system at the target product concentration.

Implementing a continuous multistage TFF operation is possible with a series of steps analogous to single-pass tangential flow filtration (SPTFF) and is well established technology [42,43]. Fig. 36.5 summarizes the concepts for continuous concentration and formulation operations. Instead of passing the product multiple times through one filter cassette in batch operation, the product is passed once through a series of filter cassettes. To achieve buffer exchange continuous cascade DF can be employed; the concentrated product stream exiting each filter cassette of the series can be combined with formulation buffer. There is significantly greater complexity in a continuous TFF than in a batch design due to the multiple membrane cassettes, which each require individual pumps, flow meters, pressure sensors, and pressure control valves. Additionally, each stage of the DF phase needs capability to dose formulation buffer at the right ratio.

An additional design consideration for continuous TFF will be the consumption requirements of formulation buffer. If every membrane stage is fed with fresh buffer, buffer consumption of a continuous system will be considerably larger than for batch operation. Frequently, formulation buffer costs contribute sizably to overall manufacturing cost due to the high grade of excipient components. Therefore, this is a disadvantage of the continuous design that must be mitigated. To maintain buffer consumption comparable to batch operation, the system has to be designed in a counter current mode, where only the last membrane stage receives fresh formulation buffer and all other stages receive the permeate of the next stage. This lowers the efficiency of buffer exchange and will increase the number of membrane stages. Therefore, system complexity and buffer consumption need to be balanced to satisfy the operational objectives of continuous processing.

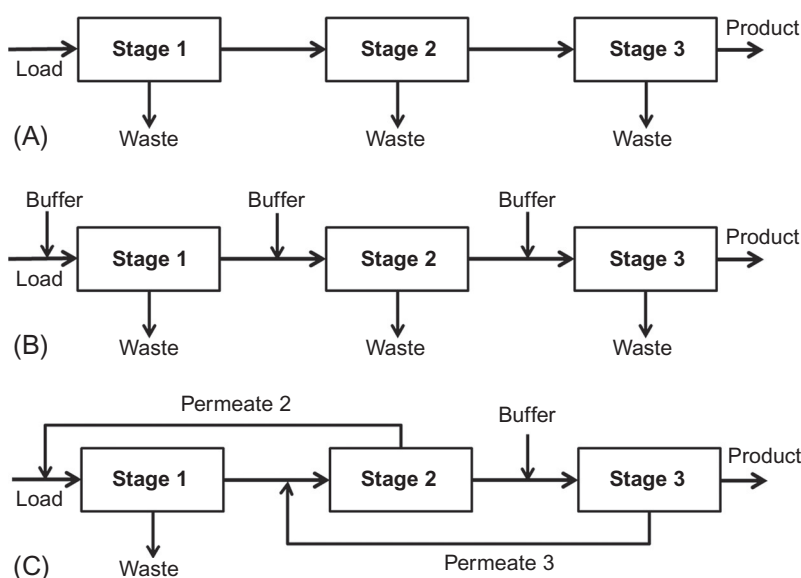


FIG. 36.5 Continuous concentration and formulation operation. (A) Continuous ultrafiltration; (B) continuous cascade diafiltration; and (C) continuous countercurrent diafiltration.

36.4 INTEGRATION OF UNIT OPERATIONS

Performing a single continuous or semicontinuous unit operation in an otherwise batch process will not provide the full benefit of continuous operations. On the other hand, integrating a series of five or more continuous or semicontinuous operations is not a trivial undertaking. Many of the operations, especially the chromatography steps, do not have a continuous steady state output streams and therefore need to be followed by an intermediate collection vessel (break tank) to create a continuous feed for the next operation. Even operations with a continuous output stream will likely also need an intermediate vessel to create a separation of unit operations to allow for flow and pressure control. This indicates that a fully integrated continuous process will be sequence of operation—tank—operation (Fig. 36.6). Process automation needs to coordinate the process flow rates of neighboring operations and the liquid level in their corresponding break tanks.

36.4.1 Bioburden Control

The combination of process complexity, and extended duration of continuous processing, as well as and the inherently growth promoting nature of most process streams in biologics manufacturing, necessitates a clear and robust strategy for bioburden control. ICH guideline Q11 on Development and Manufacture of Drug Substance [44] clearly states that, due to inherent limitations to detect low level contaminations in the drug substance, end product testing is considered inadequate and as a result additional test points have to be included in the control strategy. Current regulatory expectation is that bioburden control must be inherent to all process steps without relying on downstream sterilizing grade filtration to mitigate contamination risk. To that end, appropriate testing and bioburden control monitoring is expected before any sterilizing grade filtration. While in a continuous process no real pool may exist, an intermediate pool vessel (break tank) could transform any minor bioburden problem into a major problem by acting as reactor capable of rapidly amplifying growth. Consequently, it is expected that regulators will demand rigorous control, testing, and mitigation strategies to overcome these concerns. In principle, individual unit operations can be implemented with the same sanitization control strategies as in batch operation, but the connections between unit operations must be re-evaluated, as these lines and vessels cannot be sanitized without process interruption or line redundancy during continuous operation. The system design needs to consider clean-in-place (CIP) and steam-in-place (SIP) capabilities, equipment and line redundancies, and the ability to interrupt the continuous process for partial interim sanitization. Although these control strategies are necessary, inclusion of these features will further increase complexity, which in itself can create process vulnerability. One strategy that can minimize process and regulatory risks may be frequently scheduled process interruptions to allow intermediate preventive sanitization to sections of the process flow. This strategy comes at the cost of decreasing process efficiency, but if synchronized with additional unit changes that also pose elevated contamination risks (i.e., replacing filters), the negatives could be minimized.

36.4.2 Equipment and Plant Utilization

Both plant and equipment utilization takes on different meanings in continuous operation compared to batch processing. In batch operation utilization is expressed as the fraction of time equipment or a plant is operating. Even though equipment is frequently idle during batch production, when it is in use, the equipment is operating at its maximum design capacity. Therefore, it is easy to equate capacity utilization to the duration of operations. However, in continuous operation, plant and equipment are running 100% of the time and if using batch standards, there is an expectation of 100% utilization. Instead, utilization of continuous processes should be expressed as percent process output versus 100% process design capacity.

Running equipment 100% of the time poses additional challenges for equipment reliability and maintenance. In batch operation, the equipment frequently sits idle enabling equipment maintenance without interfering with the production schedule.

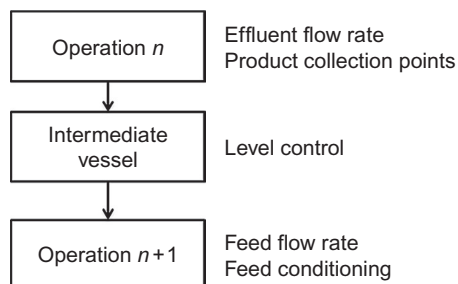


FIG. 36.6 Outline of process controls for a sequence of two connected continuous operations.

However, in continuous operation, there is no idle time for the equipment and any form of maintenance interferes with the production process. As the process is rarely running at the maximum design capability, likely there is a way to collect a process intermediate stream, perform required maintenance, and recover and catch up with the normal schedule. This adds more complexity to process control and definition though. Procedural controls should be in place for any maintenance eventuality and appropriate process responses. If these procedural controls are not in place a priori, the default response will be loss of the product stream when the process is interrupted. Additional controls will likely only be developed over time but a sophisticated regulatory strategy needs to be in place to allow adding procedural mitigations without affecting regulatory expectations and commitments. This must rely heavily on strong process characterization data and a robust process design space.

Continuous operation promises smaller equipment sizes and plant footprint due to the 24/7 operation but potentially creates other inefficiencies. In the continuous process, all operations are being performed in parallel always. This requires a completely different labor support model and documentation system for the manufacturing floor. Current practice for batch manufacturing is operator attended process execution, where operations are typically executed sequentially by the same operators. In the continuous operating model, multiple operations are performed uninterrupted and in parallel. The same team of operators that usually would move from downstream unit operation to downstream unit operation over the course of the process must support all operations at the same time. If operations are performed operator attended similarly to typical batch operation, labor costs are expected to increase. To overcome this, the continuous process needs to be executed largely unattended to be cost efficient. This requires more sophisticated, operator independent automation and automated fault detection.

For equipment design of an integrated continuous process, the turndown of process flow rates poses challenges of equipment scale. Considering the large differences in product concentration from production reactor to typical in-process streams to the final drug substance, flow rates can span a range of 10- to 50-fold difference between individual unit operations. In batch process designs, these differences are typically addressed by operating equipment of similar design and scale but varying number of process cycles for individual operations. For an integrated continuous process, equipment sizing and design for each unit operation may have to be specific to the intended scale to be able to operate at very different required process flow rates. Even though continuous processing equipment is typically smaller, if each unit operation is designed and scaled uniquely to maximize its individual capacity, this will pose as a challenge to manufacturing and engineering staff that must support multiple different sizes and designs. The protein A capture operation, the virus inactivation step, the polishing steps, the virus filtration, and the countercurrent formulation step are all operating simultaneously, each of them producing relatively small amount of product per time. For efficient operation, current manufacturing automation strategies, labor support models, and documentation requirements must be radically upgraded to allow supporting these multiple parallel operations.

36.4.3 Fault Recovery

As highlighted in the previous section, the 24/7 occupation of equipment poses new challenges when responding to process or equipment failures. One important characteristic of fault responses within the highly regulated environment of manufacturing of biologics is that any quality system expects a well-controlled process and any process failure to be analyzed and understood before further processing can commence. In current batch production practice, there is a regulatory expectation that any deviation from procedures or process outcomes is documented, the root-cause for the deviation is analyzed and detected, and a corrective and preventive action is in place before further processing [19]. Fundamentally, it is considered unacceptable to process at risk, reject failing material, and release material that passes specifications for patient use. The process is expected to always be in control.

This paradigm poses a challenge for continuous processing. In continuous processing, there is an expectation that when material deviates from quality specifications, it can be rejected, the process corrected back into normal quality range, and any material subsequently produced is deemed acceptable. Under a requirement of resolution and corrective and preventive action, any material produced before addressing a deviation may have to be quarantined in the meantime, posing operational complexities of segregation and documentation. This highlights the importance of preemptively developing wide operational design spaces and a wide range of procedural responses to anticipated deviations.

The simplest responses are responses to equipment failure. The root-cause is easily determined and mitigations are usually clear. Dependent on the perceived risk, manufacturing objectives, and quality systems mitigation approaches can vary. If the risk of failure is considered unacceptable, redundant equipment can be built into the system design from the start and SOPs for fault detection and unit switching can be put in place during system design. This leads to high reliability but low equipment utilization, as there is always an idle unit, as well as higher complexity of equipment and documentation.

For other risks, there may be a decision to divert the product stream to waste until the situation has been resolved. To prevent product loss while an investigation is ongoing, the product stream upstream of the failure point could also be collected at a stable hold point with the expectation to increase process throughput to its nominal maximum after fault correction until the accumulated product intermediate has been processed.

Responses to a process failure usually pose more difficulties. While similar considerations apply, there is a larger concern about detectability and protecting the stream of acceptable product from implicated failed product. Fast PAT will be required to immediately detect deviations and divert product to waste. Additionally, mixing behavior and residence time distributions of the entire series of operations needs to be understood and documented to enable robust product protection. For continuous processing, by the time a system can react to a detected deviation, the product stream has likely been processed over the next unit operation(s) and a clear distinction between implicated and acceptable product can be murky. Clear guidelines and capabilities to quarantine or reject implicated product must be in place to avoid implicating the entire production lot.

One of the biggest risks is bioburden contamination, primarily because the real-time detectability is low. Bioburden test results are usually only available a few days after sample submission and a contamination can have devastating impact on the entire process stream and potentially collected product from many days back. This requires rigorous qualification of effective bioburden control over the proposed process durations and beyond. Vigilant monitoring is crucial but unfortunately unlikely to be usable as part of the risk mitigation strategy. Due to its slow response, its utility is likely limited to verification of control.

36.4.4 Disposable Equipment and Devices

Continuous operation lowers volumes of in-process pools and therefore the required tank and pump sizes. Consequently, there is an expectation that the scale of operations is amenable for the use of disposable equipment even at commercial production scale [45]. Although this is fundamentally true, several aspects need to be considered. Stainless steel equipment has been used in bioprocessing for many years. Therefore, equipment design, reliability, maintenance, and cleaning are well understood. For disposable equipment, designs and reliability are not yet well established and may add significant risk to continuous operation, where equipment failure and fault recovery have potentially larger process impact. Additionally, consideration should be given to automation versus manual operator interference. A stainless-steel setup is typically designed for CIP and SIP, where all aspects of nonprocessing and equipment turnaround between processing can be automated. With disposable setups, there is a requirement for repeated manual installations of vessels and manual connections of tubing. This adds risk of connection errors, which could lead to contamination, and adds manual operator labor requirements that are very unfavorable for continuous operation as discussed before. The pressure to use disposable equipment as long as possible to minimize cost of raw material and setup time adds additional stresses for the robustness of the disposable setups. On the other hand, with a stainless-steel setup, all connections can be made by automation and their correctness can be validated. Clearly the stainless-steel setup has substantially larger capital requirements and considerably longer lead time for facility and equipment build. However, the big advantage of disposable equipment, shifting fixed cost to variable cost to ease the burden for low facility utilization situations, does not apply to continuous processing as the expectation is high facility utilization. It does not seem sensible to implement a continuous process for short duration production and low facility utilization scenarios. Consequently, there does not appear to be a significant driver to apply disposable equipment to continuous productions.

Disposable devices like prepacked chromatography columns, as well as membrane and filter cartridges are likely of more interest for continuous processing. If columns have very short bed height and are cycled frequently in a continuous column cycling process, they reach their end of lifetime rapidly, which requires many columns and membranes to be replaced on an ongoing basis. This adds significant labor burden for column packing. The use of prepacked columns can outsource that labor cost. There will be significant cost pressure on vendors to decrease premiums for packing and hardware over the resin cost, as short beds for continuous cycling use small amounts of resin, which may make use of prepacked columns cost prohibitive compared to in-house packing labor.

36.4.5 Right-Sizing Batch Size and Lot Size

One advantage of a continuous process is that batch size can be altered by extending production duration. This adds a degree of freedom analogous to producing more batches. For continuous production, it poses the challenge to define upfront a maximum batch size, as this will be the required batch size to demonstrate during PPQ. While for long term production, an extended batch duration is desirable, during PPQ however, this is a disadvantage. A PPQ campaign of a lengthy

continuous process can occupy the facility for a significant duration, and will produce a large quantity of material at a time when the commercial use of the product is not yet approved. This requires careful balance planning PPQ cost, as well as balancing the mass of product produced for inventory, and consideration of the products stability profile.

Another advantage of continuous production is the possibility to split one batch into multiple lots for testing and lot release. This allows companies to optimize lot size based on the balance between the business risk they are willing to undertake in the event of a lot failure and the costs of lot release and testing multiple batches.

Added attention must be given to raw material demand. While a continuous manufacturing process can be expected to use smaller equipment, lower flow rates, and be performed in smaller facilities, the raw material demand per mass produced is expected to be at best unchanged and may increase due to perfusion media consumption. While buffer volumes to perform a single column cycle are small as columns are small, multiple cycles must be performed to produce the same mass, therefore buffer volumes per mass produced are comparable [30]. In addition, water consumption per product mass produced will also be largely unchanged. This requires continuous processing plants to have equally large warehouse space, logistics, and buffer and media preparation capabilities as for a batch facility of equal production capacity.

36.5 SUMMARY

A fully integrated continuous process to produce therapeutic proteins is possible. Its implementation will not be trivial and requires careful planning and close collaboration of process development, manufacturing, quality, and regulatory experts. Converting the typical batch operations utilized in the production of therapeutic biologics will likely affect several established business processes and their interaction. Impacted areas include

- Process development resources and timelines.
- Labor and automation requirements on the manufacturing floor.
- Quality systems and regulatory filing strategies due to the need for different process control strategies.
- Methods for real-time detection of and response to process deviations.
- Procedures for recovery from process and equipment failures.

While each individual unit operation can be converted to continuous operation, the greatest economic benefit is expected from the integration of multiple continuous operations. Despite all these theoretical and aspirational desires however, developing a continuous process that is more cost efficient than a traditional batch process is not trivial. It will require changes to traditional automation and control strategies, to labor models on the manufacturing floor, and to overall quality systems with respect to deviation investigations and corrective and preventive actions.

From a process development perspective, focus must be on understanding where a wide design space is needed to ensure robust processing within the range of expected process conditions. As the linkage of continuous operations locks the process duration, the process execution must be able to respond to variations in product mass and concentration or stream volume by varying stream flow rates or cycle switch points within the defined process parameter ranges and expected performance indicator ranges.

From an operational perspective, focus must be on automation and control of a complex series of operations that are complex even when performed individually. Installation, operational, and performance qualification of the physical implementation of the process must be planned and executed with great attention and precision.

From a quality and regulatory perspective, substantial challenges must be expected around developing a robust strategy for bioburden control and developing responses to process deviations and equipment failures. Due to the novelty of integrated continuous processing for biologics, these aspects will be scrutinized closely by regulatory agencies during a process review to ensure product safety.

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